

Stage 3 Pilot Analysis — Sponsorship Network

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Overview

This is the pilot analysis for **Stage 3** of the Sponsorship Network experiment (Study 698e27db580428e78f8e9ff1). Stage 3 evaluators see endorsements from real Stage 2 endorsers, then stake a portion of a \$2 bonus on whether the endorser was correct.

Design: 2 x 2 x 2 between-subjects factorial

- **Endorser Gender** (Man vs. Woman)
- **Endorser Correctness** (Correct vs. Incorrect on Q1)
- **Endorsement Strength** (Strong vs. Weak — distance from neutral 50)

Primary DV: Stake percentage (0–100%) on endorser being correct

Research Questions:

RQ	Question	DV
RQ1	Are female sponsors punished more for failures / rewarded less for successes?	Trust Change (Q2 - Q1)
RQ2	Do female sponsors provide weaker endorsements?	Abs. Endorsement Strength
RQ3	Do female sponsors need higher strength to achieve same initial trust?	Q1 Stake
RQ4	Is trust change more contingent on strength x outcome for females?	Trust Change (Q2 - Q1)

1. Data Loading & Cleaning

1.1 Load Raw Export

```
raw <- read.csv("../output/stage3_pilot_data.csv", stringsAsFactors = FALSE)

# Qualtrics exports 3 header rows: labels (row 1), descriptions (row 2), import IDs (row 3)
header_labels <- raw[1, ]
raw <- raw[-c(1, 2), ] # drop label + importId rows

# Convert numeric fields
numeric_cols <- c("Progress", "Duration..in.seconds.", "Finished",
                  "endorser_slider_value_q1", "endorser_slider_value_q2",
                  "stake_percent_q1", "stake_percent_q2",
                  "rt_ms_stage3_q1", "rt_ms_stage3_q2", "rt_ms_stage3_outcome",
                  "stage3_q1_is_correct", "stage3_q1_actual_bonus",
                  "age", "gender", "race")

for (col in numeric_cols) {
  if (col %in% names(raw)) {
    raw[[col]] <- suppressWarnings(as.numeric(raw[[col]]))
  }
}

raw$StartDate <- as.POSIXct(raw$StartDate, format = "%Y-%m-%d %H:%M:%S")
raw$EndDate <- as.POSIXct(raw$EndDate, format = "%Y-%m-%d %H:%M:%S")

cat("Total rows in export:", nrow(raw), "\n")
```

```
## Total rows in export: 138
```

1.2 Exclusion Cascade

Participants must pass ALL of the following filters:

1. **Not a preview** (Status != 1)
2. **Completed survey** (Finished == 1)
3. **Valid Prolific participant** (PROLIFIC_PID is non-empty)
4. **Passed comprehension check** (Q42 == 2: correctly identified endorser information access)
5. **Has condition assignment** (JS ran successfully, assigned_condition non-empty)
6. **Has Q1 stake data** (stake_percent_q1 recorded)

```
# Build exclusion funnel
funnel <- data.frame(
  Step = character(),
  Remaining = integer(),
  Excluded = integer(),
  stringsAsFactors = FALSE
)
```

```

add_step <- function(funnel, name, data) {
  prev_n <- if (nrow(funnel) == 0) nrow(raw) else funnel$Remaining[nrow(funnel)]
  funnel <- rbind(funnel, data.frame(
    Step = name,
    Remaining = nrow(data),
    Excluded = prev_n - nrow(data),
    stringsAsFactors = FALSE
  ))
  return(funnel)
}

d0 <- raw
funnel <- add_step(funnel, "Total export rows", d0)

d1 <- d0 %>% filter(Status != "1")
funnel <- add_step(funnel, "Exclude preview responses", d1)

d2 <- d1 %>% filter(Finished == 1)
funnel <- add_step(funnel, "Exclude incomplete", d2)

d3 <- d2 %>% filter(!is.na(PROLIFIC_PID) & trimws(PROLIFIC_PID) != "")
funnel <- add_step(funnel, "Require valid PROLIFIC_PID", d3)

d3b <- d3 %>% filter(STUDY_ID == TARGET_STUDY)
funnel <- add_step(funnel, paste0("Match target study (", TARGET_STUDY, ")"), d3b)

d4 <- d3b %>% filter(Q42 == "2")
funnel <- add_step(funnel, "Pass comprehension check (Q42=2)", d4)

d5 <- d4 %>% filter(!is.na(assigned_condition) & trimws(assigned_condition) != "")
funnel <- add_step(funnel, "Has condition assignment (JS ran)", d5)

d6 <- d5 %>% filter(!is.na(stake_percent_q1))
funnel <- add_step(funnel, "Has Q1 stake data", d6)

df <- d6

styled_kable(funnel, col.names = c("Filter Step", "N Remaining", "N Excluded"))

```

Filter Step	N Remaining	N Excluded
Total export rows	138	0
Exclude preview responses	133	5
Exclude incomplete	132	1
Require valid PROLIFIC_PID	130	2
Match target study (698e27db580428e78f8e9ff1)	95	35
Pass comprehension check (Q42=2)	42	53
Has condition assignment (JS ran)	42	0
Has Q1 stake data	42	0

```

cat("\nFinal analytic sample: N =", nrow(df), "\n")

```

```
##
```

```
## Final analytic sample: N = 42
```

1.3 Derive Variables

```
df <- df %>%
  mutate(
    # Extract factors from assigned_condition (e.g., "M_correct_strong")
    endorser_gender_code = ifelse(grepl("^M_", assigned_condition), "Man", "Woman"),
    endorser_correct      = ifelse(grepl("_correct_", assigned_condition), 1, 0),
    endorser_strong       = ifelse(grepl("_strong$", assigned_condition), 1, 0),

    # Compute absolute endorsement strength (distance from neutral 50)
    abs_strength_q1 = abs(endorser_slider_value_q1 - 50),
    abs_strength_q2 = abs(endorser_slider_value_q2 - 50),

    # Factor versions for modeling
    endorser_gender_f = factor(endorser_gender_code, levels = c("Man", "Woman")),
    correctness_f      = factor(ifelse(endorser_correct == 1, "Correct", "Incorrect"),
                                levels = c("Incorrect", "Correct")),
    strength_f         = factor(ifelse(endorser_strong == 1, "Strong", "Weak"),
                                levels = c("Weak", "Strong")),
    outcome_f          = factor(ifelse(stage3_q1_is_correct == 1, "Success", "Failure"),
                                levels = c("Failure", "Success")),

    # Trust change = Q2 stake - Q1 stake
    trust_change = stake_percent_q2 - stake_percent_q1,

    # Participant demographics
    participant_gender = case_when(
      gender == 1 ~ "Male",
      gender == 2 ~ "Female",
      gender == 3 ~ "Other/Non-binary",
      TRUE ~ NA_character_
    ),

    # Duration in minutes
    duration_min = Duration..in.seconds. / 60
  )

n_q2      <- sum(!is.na(df$trust_change))
n_rt      <- sum(df$rt_ms_stage3_q1 > 0, na.rm = TRUE)
n_conds   <- length(unique(df$assigned_condition))

cat("Derived variables created.\n")
```

```
## Derived variables created.
```

```
cat("  Total valid N:", nrow(df), "\n")
```

```
##   Total valid N: 42
```

```
cat("  Trust change computable (has Q2):", n_q2, "/", nrow(df), "\n")
```

```
##  Trust change computable (has Q2): 42 / 42
```

```
cat("  Response times recorded:", n_rt, "/", nrow(df), "\n")
```

```
##  Response times recorded: 42 / 42
```

```
cat("  Conditions covered:", n_conds, "/ 8\n")
```

```
##  Conditions covered: 8 / 8
```

2. Data Quality & Diagnostics

2.1 Response Overview

```
overview <- data.frame(
  Metric = c(
    "Total export rows",
    "Final analytic sample",
    "Q1 stake recorded",
    "Q2 stake recorded",
    "Trust change computable",
    "Response times recorded (Q1)",
    "Conditions covered (of 8)",
    paste0("From target study (", TARGET_STUDY, ")")
  ),
  Value = c(
    nrow(raw),
    nrow(df),
    sum(!is.na(df$stake_percent_q1)),
    sum(!is.na(df$stake_percent_q2)),
    n_q2,
    n_rt,
    n_conds,
    sum(df$STUDY_ID == TARGET_STUDY, na.rm = TRUE)
  )
)

styled_kable(overview, align = "lr")
```

Metric	Value
Total export rows	138
Final analytic sample	42
Q1 stake recorded	42
Q2 stake recorded	42
Trust change computable	42
Response times recorded (Q1)	42
Conditions covered (of 8)	8
From target study (698e27db580428e78f8e9ff1)	42

2.2 Comprehension Check Breakdown

Q42 asks: “When the endorsers made their predictions, what information did they have access to?”

- Choice 1: “Only the candidates’ logical-reasoning test scores”
- **Choice 2 (CORRECT): “The candidates’ knowledge quiz and/or word-search puzzle scores (some may have been listed as ‘Unknown’)”**
- Choice 3: “All three task scores: logical-reasoning, knowledge quiz, and word-search”
- Choice 4: “The candidates’ names and demographics”

This is a **hard gate** in the survey — incorrect answers end the survey with redirect code C100PIGA. We still document it here for transparency.

```
# Among all Prolific participants who completed the survey
comp_pool <- raw %>%
  filter(Status != "1", Finished == 1,
         !is.na(PROLIFIC_PID), trimws(PROLIFIC_PID) != "")

comp_tbl <- comp_pool %>%
  count(Q42) %>%
  mutate(
    Label = case_when(
      Q42 == "1" ~ "Only logical-reasoning scores",
      Q42 == "2" ~ "Knowledge quiz / word-search (CORRECT)",
      Q42 == "3" ~ "All three task scores",
      Q42 == "4" ~ "Names and demographics",
      TRUE ~ "No response / screened before Q42"
    ),
    Pct = paste0(round(n / sum(n) * 100, 1), "%")
  )

styled_kable(comp_tbl, col.names = c("Q42 Value", "Label", "N", "%"))
```

Q42 Value	Label	N	%
	11	No response / screened before Q42	8.5%
1	13	Only logical-reasoning scores	10%
2	65	Knowledge quiz / word-search (CORRECT)	50%
3	39	All three task scores	30%
4	2	Names and demographics	1.5%

2.3 Attention Checks (Effort Checks)

```
# ec_1 items: bee sting, Thailand, Venus, water, fatal heart attack, human
# Trap items with objectively correct answers:
#   Venus = 2 (No), water = 1 (Yes), fatal heart attack = 2 (No), human = 1 (Yes)

df <- df %>%
  mutate(
    check_venus = (ec_1_3 == "2"),
    check_water = (ec_1_4 == "1"),
    check_fatal = (ec_1_5 == "2"),
    check_human = (ec_1_6 == "1"),
    check_text = grepl("^\\\\"?one two\\\\"?$", tolower(trimws(ec_2))),
    attn_passed = check_venus & check_water & check_fatal & check_human & check_text,
    n_checks_passed = check_venus + check_water + check_fatal + check_human + check_text
  )

attn_summary <- data.frame(
  Check = c("Venus (never been)", "Water (has drunk)", "Fatal heart attack (no)",
            "Human (yes)", "Text entry ('one two')", "ALL passed"),
```

```

Pass_N = c(sum(df$check_venus, na.rm = TRUE),
            sum(df$check_water, na.rm = TRUE),
            sum(df$check_fatal, na.rm = TRUE),
            sum(df$check_human, na.rm = TRUE),
            sum(df$check_text, na.rm = TRUE),
            sum(df$attn_passed, na.rm = TRUE)),
Pass_Pct = paste0(round(c(
  mean(df$check_venus, na.rm = TRUE),
  mean(df$check_water, na.rm = TRUE),
  mean(df$check_fatal, na.rm = TRUE),
  mean(df$check_human, na.rm = TRUE),
  mean(df$check_text, na.rm = TRUE),
  mean(df$attn_passed, na.rm = TRUE)
) * 100, 1), "%")
)

styled_kable(attn_summary, col.names = c("Check", "N Passed", "% Passed"),
  caption = paste0("Attention checks among final sample (N=", nrow(df), ")"))

```

Table 5: Attention checks among final sample (N=42)

Check	N Passed	% Passed
Venus (never been)	42	100%
Water (has drunk)	42	100%
Fatal heart attack (no)	42	100%
Human (yes)	42	100%
Text entry ('one two')	42	100%
ALL passed	42	100%

```

cat("\nAttention check failures in final sample:", sum(!df$attn_passed, na.rm = TRUE), "\n")

```

```

##
## Attention check failures in final sample: 0

```

2.4 Manipulation Check: Endorser Gender

```

# Q44: "What is the gender of the endorser?" (1=Man, 2=Woman, 3=I don't remember)
df <- df %>%
  mutate(
    q44_response = case_when(
      Q44 == "1" ~ "Man",
      Q44 == "2" ~ "Woman",
      Q44 == "3" ~ "Don't remember",
      TRUE ~ NA_character_
    ),
    gender_manip_correct = (q44_response == endorser_gender_code)
  )

manip_tbl <- df %>%

```

```

filter(!is.na(q44_response)) %>%
group_by(endorser_gender_f) %>%
summarise(
  N = n(),
  Correct = sum(gender_manip_correct, na.rm = TRUE),
  `Don't Remember` = sum(q44_response == "Don't remember", na.rm = TRUE),
  Wrong = N - Correct - `Don't Remember`,
  Accuracy = paste0(round(Correct / N * 100, 1), "%"),
  .groups = "drop"
)

styled_kable(manip_tbl)

```

endorser_gender_f	N	Correct	Don't Remember	Wrong	Accuracy
Man	20	19	0	1	95%
Woman	19	16	0	3	84.2%

```

overall_acc <- mean(df$gender_manip_correct, na.rm = TRUE)
cat("\nOverall manipulation check accuracy:", round(overall_acc * 100, 1), "%\n")

```

```

##
## Overall manipulation check accuracy: 89.7 %

```

```

if (overall_acc < 0.7) {
  cat("WARNING: Gender manipulation accuracy below 70% - consider strengthening the gender cue.\n")
}

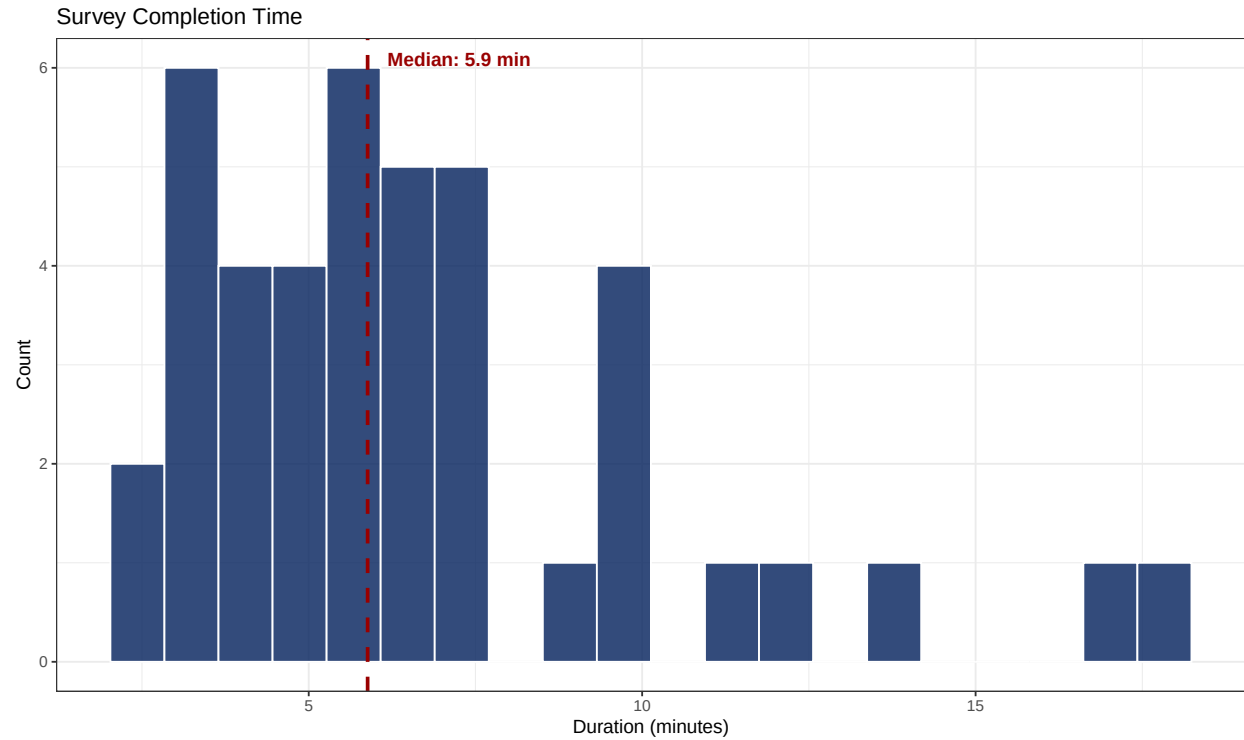
```

2.5 Completion Time

```

ggplot(df, aes(x = duration_min)) +
  geom_histogram(bins = 20, fill = PENN_BLUE, color = "white", alpha = 0.8) +
  geom_vline(xintercept = median(df$duration_min, na.rm = TRUE),
    linetype = "dashed", color = PENN_RED, size = 1) +
  annotate("text", x = median(df$duration_min, na.rm = TRUE) + 0.3,
    y = Inf, vjust = 2, hjust = 0,
    label = paste0("Median: ", round(median(df$duration_min, na.rm = TRUE), 1), " min"),
    color = PENN_RED, fontface = "bold") +
  labs(title = "Survey Completion Time",
    x = "Duration (minutes)", y = "Count") +
  theme_bw()

```



```
cat("Duration stats (minutes):\n")
```

```
## Duration stats (minutes):
```

```
cat("  Mean:", round(mean(df$duration_min, na.rm = TRUE), 1), "\n")
```

```
##   Mean: 6.6
```

```
cat("  Median:", round(median(df$duration_min, na.rm = TRUE), 1), "\n")
```

```
##   Median: 5.9
```

```
cat("  SD:", round(sd(df$duration_min, na.rm = TRUE), 1), "\n")
```

```
##   SD: 3.6
```

```
cat("  Range:", round(min(df$duration_min, na.rm = TRUE), 1), "-",  
    round(max(df$duration_min, na.rm = TRUE), 1), "\n")
```

```
##   Range: 2.1 - 17.5
```

3. Sample Description

3.1 Demographics

```
cat("Age:\n")
```

```
## Age:
```

```
cat("  N:", sum(!is.na(df$age)), "\n")
```

```
##   N: 42
```

```
cat("  Mean:", round(mean(df$age, na.rm = TRUE), 1), "\n")
```

```
##   Mean: 45
```

```
cat("  SD:", round(sd(df$age, na.rm = TRUE), 1), "\n")
```

```
##   SD: 11.9
```

```
cat("  Range:", min(df$age, na.rm = TRUE), "-", max(df$age, na.rm = TRUE), "\n\n")
```

```
##   Range: 25 - 74
```

```
gender_tbl <- df %>%  
  count(participant_gender) %>%  
  mutate(Pct = paste0(round(n / sum(n) * 100, 1), "%"))  
styled_kable(gender_tbl, col.names = c("Gender", "N", "%"),  
  caption = "Participant gender")
```

Table 7: Participant gender

Gender	N	%
Female	23	54.8%
Male	17	40.5%
Other/Non-binary	2	4.8%

```
race_labels <- c("1" = "AI/AN", "2" = "Asian", "3" = "Black/AA",  
  "4" = "Hispanic/Latino", "5" = "White", "6" = "Other")  
race_tbl <- df %>%  
  mutate(race_label = ifelse(!is.na(race), race_labels[as.character(race)], "Missing")) %>%  
  count(race_label) %>%  
  mutate(Pct = paste0(round(n / sum(n) * 100, 1), "%"))  
styled_kable(race_tbl, col.names = c("Race/Ethnicity", "N", "%"))
```

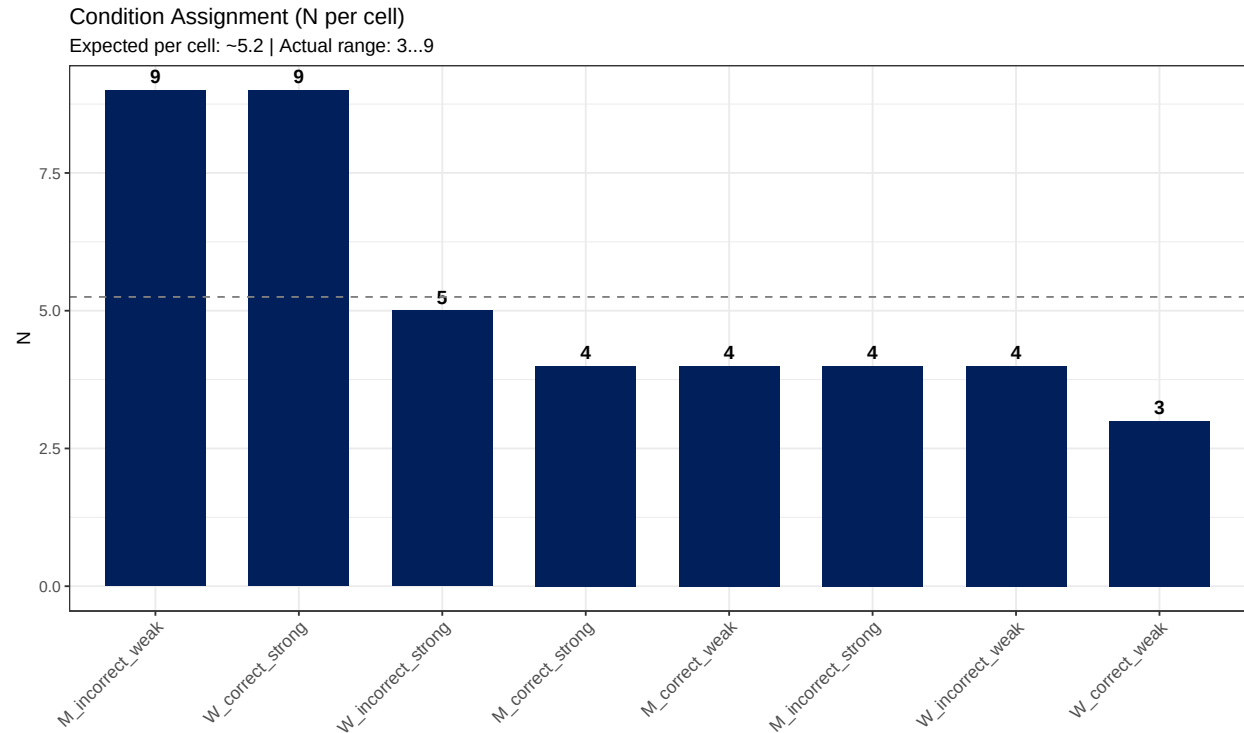
Race/Ethnicity	N	%
Asian	5	11.9%
Black/AA	5	11.9%
White	32	76.2%

3.2 Condition Assignment

```
cond_tbl <- df %>%
  count(assigned_condition) %>%
  arrange(assigned_condition)

all_conds <- c("M_correct_strong", "M_correct_weak", "M_incorrect_strong",
              "M_incorrect_weak", "W_correct_strong", "W_correct_weak",
              "W_incorrect_strong", "W_incorrect_weak")
cond_tbl <- data.frame(assigned_condition = all_conds) %>%
  left_join(cond_tbl, by = "assigned_condition") %>%
  mutate(n = ifelse(is.na(n), 0, n))

ggplot(cond_tbl, aes(x = reorder(assigned_condition, -n), y = n,
                        fill = ifelse(n == 0, "Empty", "Filled"))) +
  geom_bar(stat = "identity", width = 0.7) +
  geom_text(aes(label = n), vjust = -0.5, fontface = "bold") +
  geom_hline(yintercept = nrow(df) / 8, linetype = "dashed", color = "gray50") +
  scale_fill_manual(values = c("Empty" = "gray80", "Filled" = PENN_BLUE), guide = "none") +
  labs(title = "Condition Assignment (N per cell)",
       subtitle = paste0("Expected per cell: ~", round(nrow(df) / 8, 1),
                        " | Actual range: ", min(cond_tbl$n), "-", max(cond_tbl$n)),
       x = NULL, y = "N") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 10))
```



```
factor_bal <- data.frame(
  Factor = c("Endorser Gender: Man", "Endorser Gender: Woman",
            "Correctness: Correct", "Correctness: Incorrect",
            "Strength: Strong", "Strength: Weak"),
  N = c(sum(df$endorser_gender_code == "Man"),
        sum(df$endorser_gender_code == "Woman"),
        sum(df$endorser_correct == 1),
        sum(df$endorser_correct == 0),
        sum(df$endorser_strong == 1),
        sum(df$endorser_strong == 0)),
  Pct = paste0(round(c(
    mean(df$endorser_gender_code == "Man"),
    mean(df$endorser_gender_code == "Woman"),
    mean(df$endorser_correct == 1),
    mean(df$endorser_correct == 0),
    mean(df$endorser_strong == 1),
    mean(df$endorser_strong == 0)
  ) * 100, 1), "%")
)

styled_kable(factor_bal, col.names = c("Factor Level", "N", "%"))
```

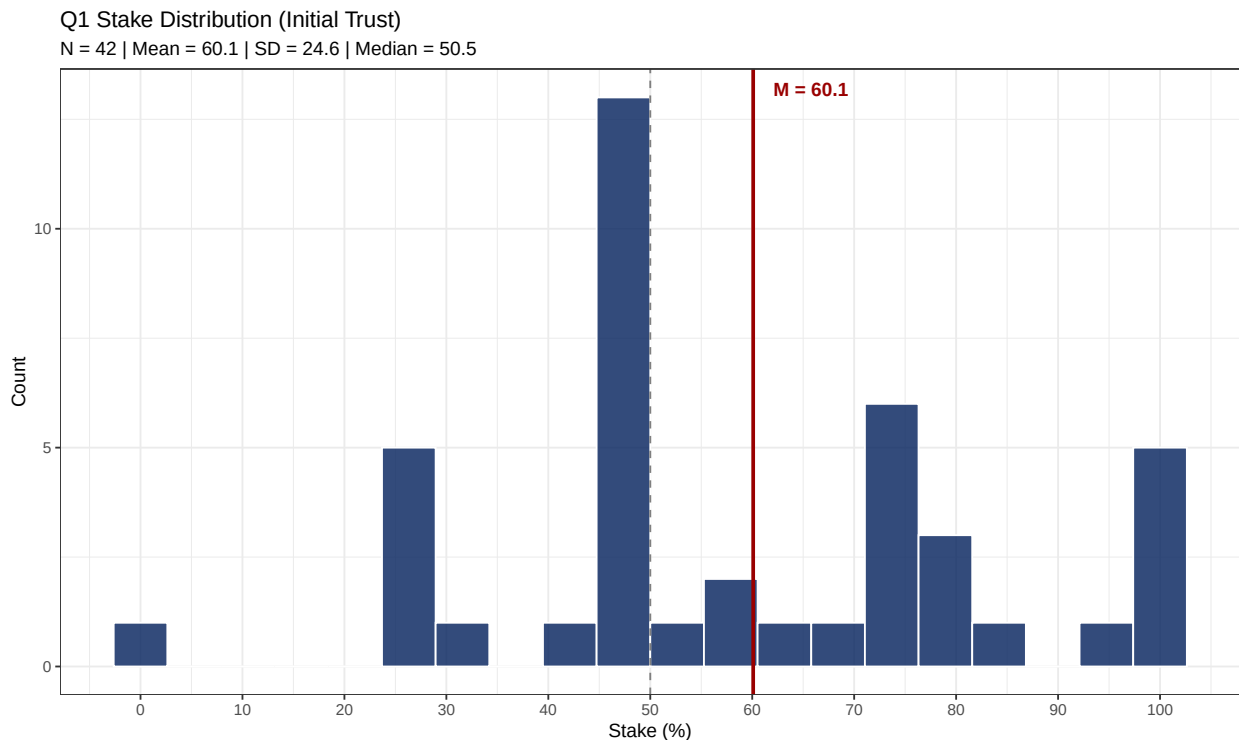
Factor Level	N	%
Endorser Gender: Man	21	50%
Endorser Gender: Woman	21	50%
Correctness: Correct	20	47.6%
Correctness: Incorrect	22	52.4%
Strength: Strong	22	52.4%

Factor Level	N	%
Strength: Weak	20	47.6%

4. Primary DVs: Q1 Stake & Trust Change

4.1 Q1 Stake Distribution (Initial Trust)

```
ggplot(df, aes(x = stake_percent_q1)) +  
  geom_histogram(bins = 20, fill = PENN_BLUE, color = "white", alpha = 0.8) +  
  geom_vline(xintercept = 50, linetype = "dashed", color = "gray50") +  
  geom_vline(xintercept = mean(df$stake_percent_q1, na.rm = TRUE),  
            linetype = "solid", color = PENN_RED, size = 1) +  
  annotate("text", x = mean(df$stake_percent_q1) + 2, y = Inf, vjust = 2, hjust = 0,  
         label = paste0("M = ", round(mean(df$stake_percent_q1), 1)),  
         color = PENN_RED, fontface = "bold") +  
  scale_x_continuous(breaks = seq(0, 100, 10)) +  
  labs(title = "Q1 Stake Distribution (Initial Trust)",  
       subtitle = paste0("N = ", nrow(df), " | Mean = ", round(mean(df$stake_percent_q1), 1),  
                        " | SD = ", round(sd(df$stake_percent_q1), 1),  
                        " | Median = ", median(df$stake_percent_q1)),  
       x = "Stake (%)", y = "Count") +  
  theme_bw()
```



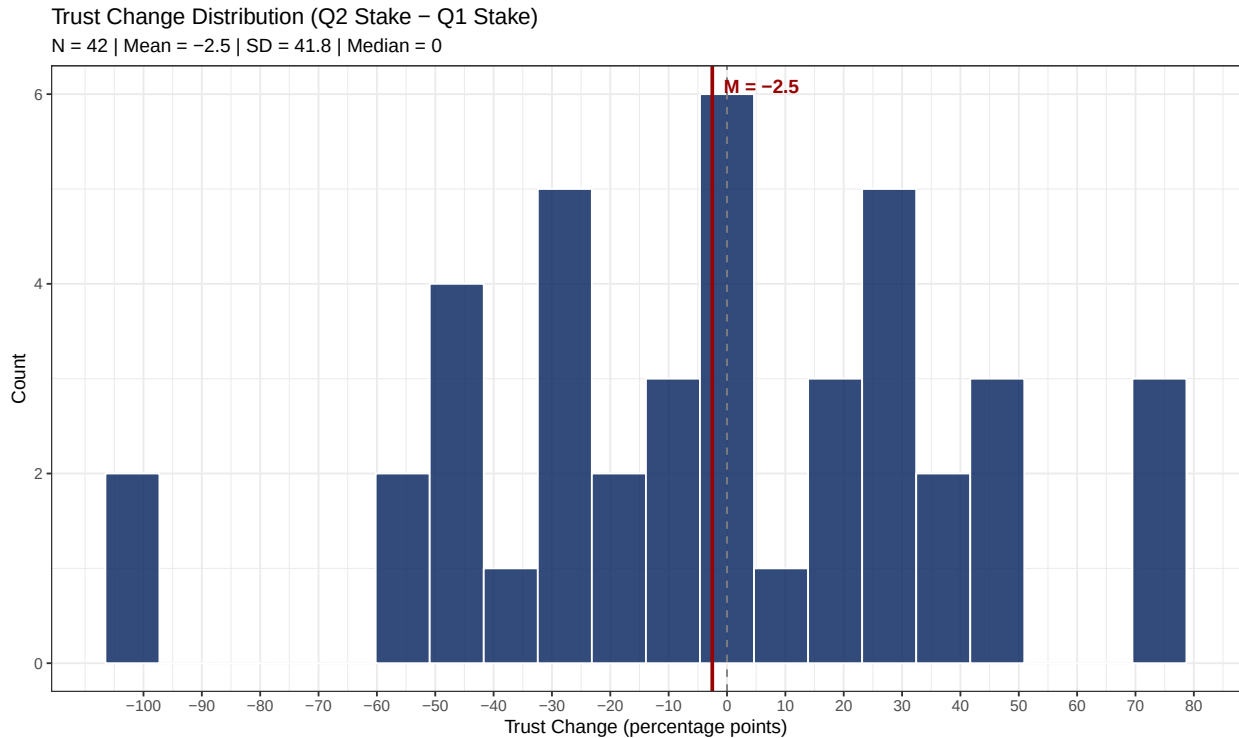
4.2 Trust Change Distribution (Q2 - Q1)

```
df_tc <- df %>% filter(!is.na(trust_change))  
  
if (nrow(df_tc) > 0) {  
  ggplot(df_tc, aes(x = trust_change)) +
```

```

geom_histogram(bins = 20, fill = PENN_BLUE, color = "white", alpha = 0.8) +
geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
geom_vline(xintercept = mean(df_tc$trust_change),
           linetype = "solid", color = PENN_RED, size = 1) +
annotate("text", x = mean(df_tc$trust_change) + 2, y = Inf, vjust = 2, hjust = 0,
         label = paste0("M = ", round(mean(df_tc$trust_change), 1)),
         color = PENN_RED, fontface = "bold") +
scale_x_continuous(breaks = seq(-100, 100, 10)) +
labs(title = "Trust Change Distribution (Q2 Stake - Q1 Stake)",
     subtitle = paste0("N = ", nrow(df_tc), " | Mean = ", round(mean(df_tc$trust_change), 1),
                       " | SD = ", round(sd(df_tc$trust_change), 1),
                       " | Median = ", median(df_tc$trust_change)),
     x = "Trust Change (percentage points)", y = "Count") +
theme_bw()
} else {
  cat("No trust change data available (Q2 stake missing for all responses).\n")
}

```



```
cat("Trust change summary:\n")
```

```
## Trust change summary:
```

```
cat("  N with Q2:", nrow(df_tc), "/", nrow(df), "\n")
```

```
##  N with Q2: 42 / 42
```

```

if (nrow(df_tc) > 0) {
  cat("  Mean:", round(mean(df_tc$trust_change), 1), "\n")
  cat("  SD:", round(sd(df_tc$trust_change), 1), "\n")
  cat("  Range:", round(min(df_tc$trust_change), 1), "to",
      round(max(df_tc$trust_change), 1), "\n")
}

```

```

##  Mean: -2.5
##  SD: 41.8
##  Range: -100 to 76

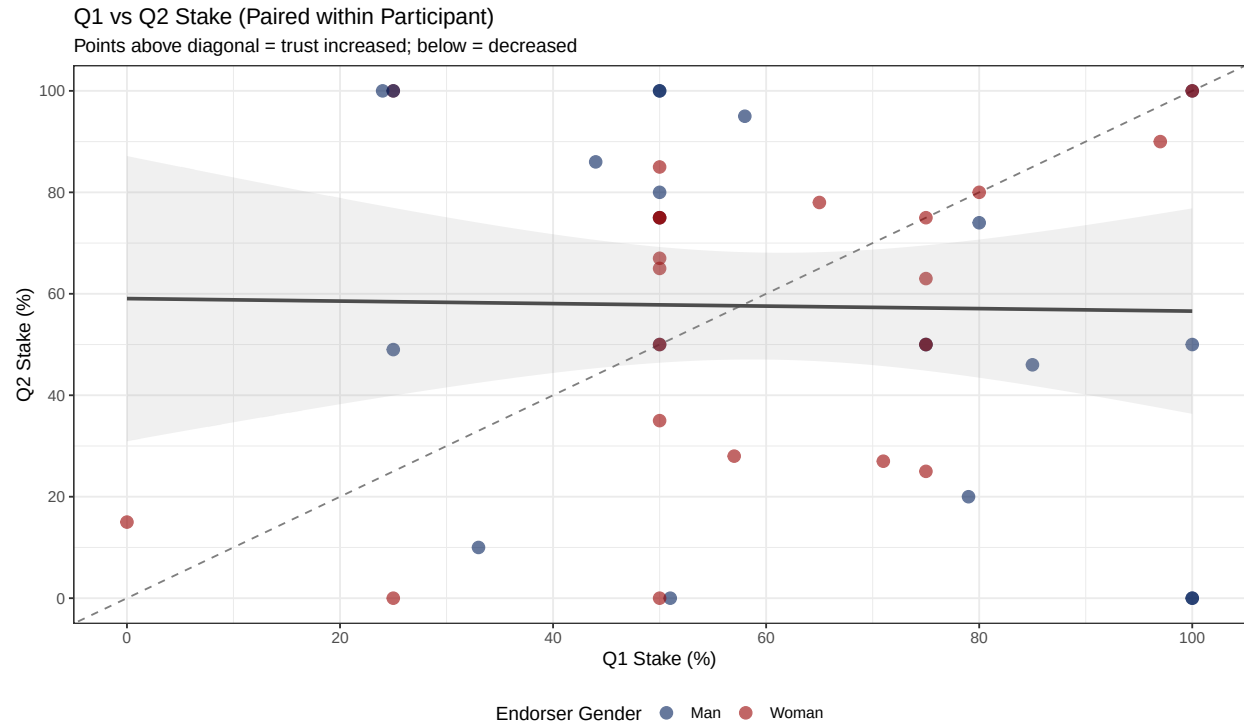
```

4.3 Q1 vs Q2 Stake (Paired)

```

if (nrow(df_tc) > 2) {
  ggplot(df_tc, aes(x = stake_percent_q1, y = stake_percent_q2)) +
    geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "gray50") +
    geom_point(aes(color = endorser_gender_f), alpha = 0.6, size = 3) +
    geom_smooth(method = "lm", se = TRUE, color = "gray30", alpha = 0.15) +
    scale_color_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED)) +
    scale_x_continuous(limits = c(0, 100), breaks = seq(0, 100, 20)) +
    scale_y_continuous(limits = c(0, 100), breaks = seq(0, 100, 20)) +
    labs(title = "Q1 vs Q2 Stake (Paired within Participant)",
         subtitle = "Points above diagonal = trust increased; below = decreased",
         x = "Q1 Stake (%)", y = "Q2 Stake (%)",
         color = "Endorser Gender") +
    theme_bw() +
    theme(legend.position = "bottom")
}

```



4.4 Stake by Condition (All Cells)

```
cond_stats <- df %>%
  group_by(assigned_condition) %>%
  summarise(
    n = n(),
    q1_mean = round(mean(stake_percent_q1), 1),
    q1_sd = round(sd(stake_percent_q1), 1),
    q2_mean = ifelse(sum(!is.na(stake_percent_q2)) > 0,
                     round(mean(stake_percent_q2, na.rm = TRUE), 1), NA),
    tc_mean = ifelse(sum(!is.na(trust_change)) > 0,
                     round(mean(trust_change, na.rm = TRUE), 1), NA),
    n_q2 = sum(!is.na(stake_percent_q2)),
    .groups = "drop"
  ) %>%
  arrange(assigned_condition)

styled_kable(cond_stats,
  col.names = c("Condition", "N", "Q1 Mean", "Q1 SD",
                "Q2 Mean", "Trust Change", "N (Q2)"))
```

Condition	N	Q1 Mean	Q1 SD	Q2 Mean	Trust Change	N (Q2)
M_correct_strong	4	70.0	24.5	87.2	17.2	4
M_correct_weak	4	37.8	14.7	62.2	24.5	4
M_incorrect_strong	4	78.5	4.7	41.5	-37.0	4
M_incorrect_weak	9	62.1	30.1	52.3	-9.8	9

Condition	N	Q1 Mean	Q1 SD	Q2 Mean	Trust Change	N (Q2)
W_correct_strong	9	66.6	16.5	60.7	-5.9	9
W_correct_weak	3	41.7	14.4	61.7	20.0	3
W_incorrect_strong	5	49.2	34.7	33.4	-15.8	5
W_incorrect_weak	4	62.5	25.0	71.2	8.8	4

5. RQ3: Initial Trust ~ Endorser Gender x Endorsement Strength

Hypothesis (HQ3): Female sponsors need **higher** endorsement strength to achieve the same initial trust. Predicts a negative interaction: female endorsers have a shallower slope relating strength to trust.

$$\text{Stake}_{Q1} = \beta_0 + \beta_1 \text{Female} + \beta_2 \text{AbsStrength} + \beta_3 (\text{Female} \times \text{AbsStrength}) + \epsilon$$

5.1 Model

```
model_rq3 <- lm(stake_percent_q1 ~ endorser_gender_f * abs_strength_q1, data = df)
summary(model_rq3)
```

```
##
## Call:
## lm(formula = stake_percent_q1 ~ endorser_gender_f * abs_strength_q1,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -60.352 -10.352  -3.562   13.718   46.438
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   54.2040     6.9473   7.802 2.08e-09 ***
## endorser_gender_fWoman          -0.7802    11.7142  -0.067  0.9473
## abs_strength_q1                  0.3974     0.2250   1.766  0.0854 .
## endorser_gender_fWoman:abs_strength_q1 -0.2588     0.3225  -0.803  0.4271
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 24.38 on 38 degrees of freedom
## Multiple R-squared:  0.09009,    Adjusted R-squared:  0.01826
## F-statistic: 1.254 on 3 and 38 DF,  p-value: 0.3038
```

```
coefs <- summary(model_rq3)$coefficients
coef_tbl <- data.frame(
  Term = rownames(coefs),
  Estimate = round(coefs[, 1], 3),
  SE = round(coefs[, 2], 3),
  t = round(coefs[, 3], 3),
  p = round(coefs[, 4], 4)
)
rownames(coef_tbl) <- NULL

coef_tbl$Sig <- ifelse(coef_tbl$p < 0.05, "*", "")

styled_kable(coef_tbl, col.names = c("Term", "Estimate", "SE", "t", "p", ""),
  caption = "RQ3 regression coefficients")
```

Table 11: RQ3 regression coefficients

Term	Estimate	SE	t	p
(Intercept)	54.204	6.947	7.802	0.0000 *
endorser_gender_fWoman	-0.780	11.714	-0.067	0.9473
abs_strength_q1	0.397	0.225	1.766	0.0854
endorser_gender_fWoman:abs_strength_q1	-0.259	0.322	-0.803	0.4271

Interpretation:

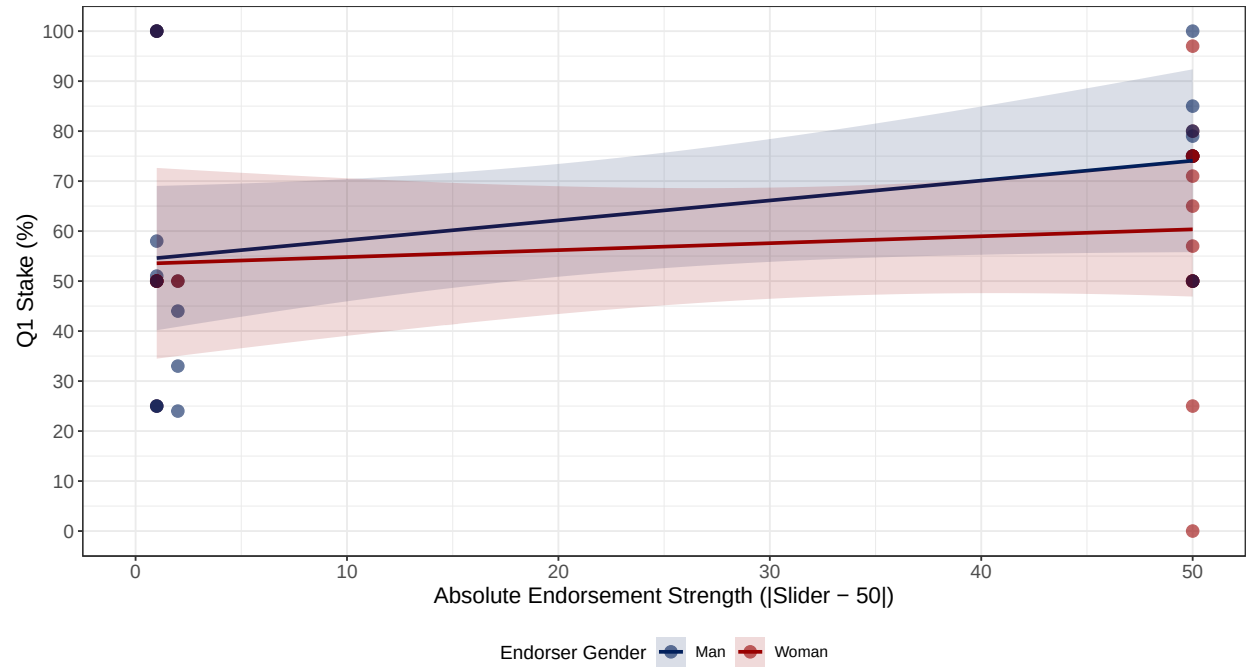
- (Intercept): Baseline stake for **male** endorsers at **zero** absolute strength
- endorser_gender_fWoman: Difference in baseline stake for women vs. men
- abs_strength_q1: Slope for males — how much stake increases per unit of endorsement strength
- endorser_gender_fWoman:abs_strength_q1: **HQ3 test** — negative = females have shallower slope

5.2 Visualization

```
ggplot(df, aes(x = abs_strength_q1, y = stake_percent_q1,
               color = endorser_gender_f, fill = endorser_gender_f)) +
  geom_point(alpha = 0.6, size = 3) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15) +
  scale_color_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED)) +
  scale_fill_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED)) +
  scale_x_continuous(breaks = seq(0, 50, 10), limits = c(0, 50)) +
  scale_y_continuous(limits = c(0, 100), breaks = seq(0, 100, 10)) +
  labs(
    title = "RQ3: Do Female Endorsers Need Higher Strength for Same Trust?",
    subtitle = paste0("Pilot (N = ", nrow(df), ") | Compare to simulation predictions"),
    x = "Absolute Endorsement Strength (|Slider - 50|)",
    y = "Q1 Stake (%)",
    color = "Endorser Gender", fill = "Endorser Gender"
  ) +
  theme_bw() +
  theme(
    legend.position = "bottom",
    plot.title = element_text(size = 14, face = "bold"),
    axis.text = element_text(size = 11),
    axis.title = element_text(size = 13)
  )
```

RQ3: Do Female Endorsers Need Higher Strength for Same Trust?

Pilot (N = 42) | Compare to simulation predictions



6. RQ1: Trust Change by Endorser Gender x Outcome

Hypothesis (HQ1): Female sponsors are punished **more** for failures and rewarded **less** for successes.

$$\text{TrustChange} = \beta_0 + \beta_1 \text{Female} + \beta_2 \text{Success} + \beta_3 (\text{Female} \times \text{Success}) + \epsilon$$

Requires Q2 stake data. Analysis restricted to participants with both Q1 and Q2.

```
cat("Participants with trust change data:", nrow(df_tc), "\n")
```

```
## Participants with trust change data: 42
```

6.1 Model

```
if (nrow(df_tc) >= 10) {
  model_rq1 <- lm(trust_change ~ endorser_gender_f * outcome_f, data = df_tc)
  summary(model_rq1)

  coefs_rq1 <- summary(model_rq1)$coefficients
  coef_tbl_rq1 <- data.frame(
    Term = rownames(coefs_rq1),
    Estimate = round(coefs_rq1[, 1], 3),
    SE = round(coefs_rq1[, 2], 3),
    t = round(coefs_rq1[, 3], 3),
    p = round(coefs_rq1[, 4], 4)
  )
  rownames(coef_tbl_rq1) <- NULL

  coef_tbl_rq1$Sig <- ifelse(coef_tbl_rq1$p < 0.05, "*", "")

  styled_kable(coef_tbl_rq1, col.names = c("Term", "Estimate", "SE", "t", "p", ""),
    caption = "RQ1 regression coefficients")
} else {
  cat("Insufficient data for RQ1 model (need >= 10 with trust change).\n")
}
```

Table 12: RQ1 regression coefficients

Term	Estimate	SE	t	p
(Intercept)	-18.154	11.384	-1.595	0.1191
endorser_gender_fWoman	13.265	17.798	0.745	0.4607
outcome_fSuccess	39.029	18.444	2.116	0.0409
endorser_gender_fWoman:outcome_fSuccess	-33.557	25.841	-1.299	0.2019

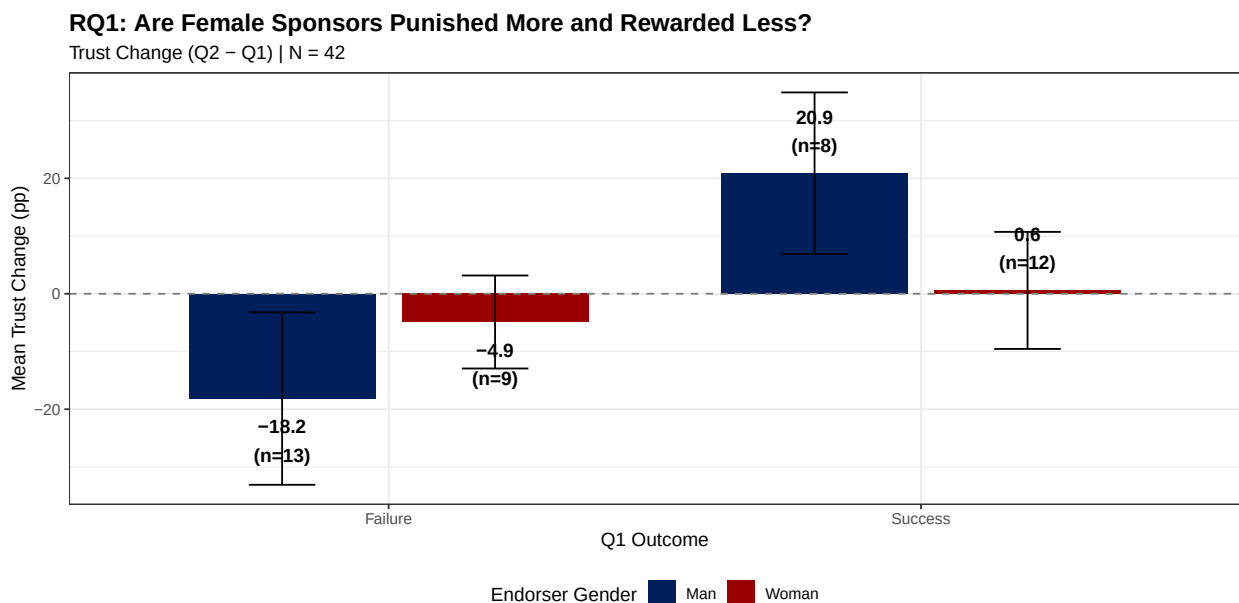
6.2 Visualization

```

if (nrow(df_tc) >= 4) {
  rq1_stats <- df_tc %>%
    group_by(endorser_gender_f, outcome_f) %>%
    summarise(
      n = n(),
      mean_tc = mean(trust_change),
      se = ifelse(n() > 1, sd(trust_change) / sqrt(n()), 0),
      .groups = "drop"
    )

  ggplot(rq1_stats, aes(x = outcome_f, y = mean_tc, fill = endorser_gender_f)) +
    geom_bar(stat = "identity", position = position_dodge(0.8), width = 0.7) +
    geom_errorbar(aes(ymin = mean_tc - se, ymax = mean_tc + se),
                  position = position_dodge(0.8), width = 0.25) +
    geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
    geom_text(aes(label = sprintf("%.1f\nn(n=%d)", mean_tc, n)),
              position = position_dodge(0.8),
              vjust = ifelse(rq1_stats$mean_tc >= 0, -0.5, 1.5),
              size = 4, fontface = "bold") +
    scale_fill_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED)) +
    labs(
      title = "RQ1: Are Female Sponsors Punished More and Rewarded Less?",
      subtitle = paste0("Trust Change (Q2 - Q1) | N = ", nrow(df_tc)),
      x = "Q1 Outcome", y = "Mean Trust Change (pp)", fill = "Endorser Gender"
    ) +
    theme_bw() +
    theme(legend.position = "bottom",
          plot.title = element_text(size = 14, face = "bold"))
}

```



7. RQ4: Trust Change ~ Gender x Strength x Outcome (3-Way)

Hypothesis (HQ4): Trust change is **more** contingent on endorsement strength x outcome for female sponsors.

$$\text{TrustChange} = \beta_0 + \beta_1 F + \beta_2 S^c + \beta_3 O + \beta_4 (F \times S^c) + \beta_5 (F \times O) + \beta_6 (S^c \times O) + \beta_7 (F \times S^c \times O) + \epsilon$$

where S^c = mean-centered absolute strength.

```
if (nrow(df_tc) >= 15) {
  df_tc <- df_tc %>%
    mutate(abs_strength_q1_c = abs_strength_q1 - mean(abs_strength_q1, na.rm = TRUE))

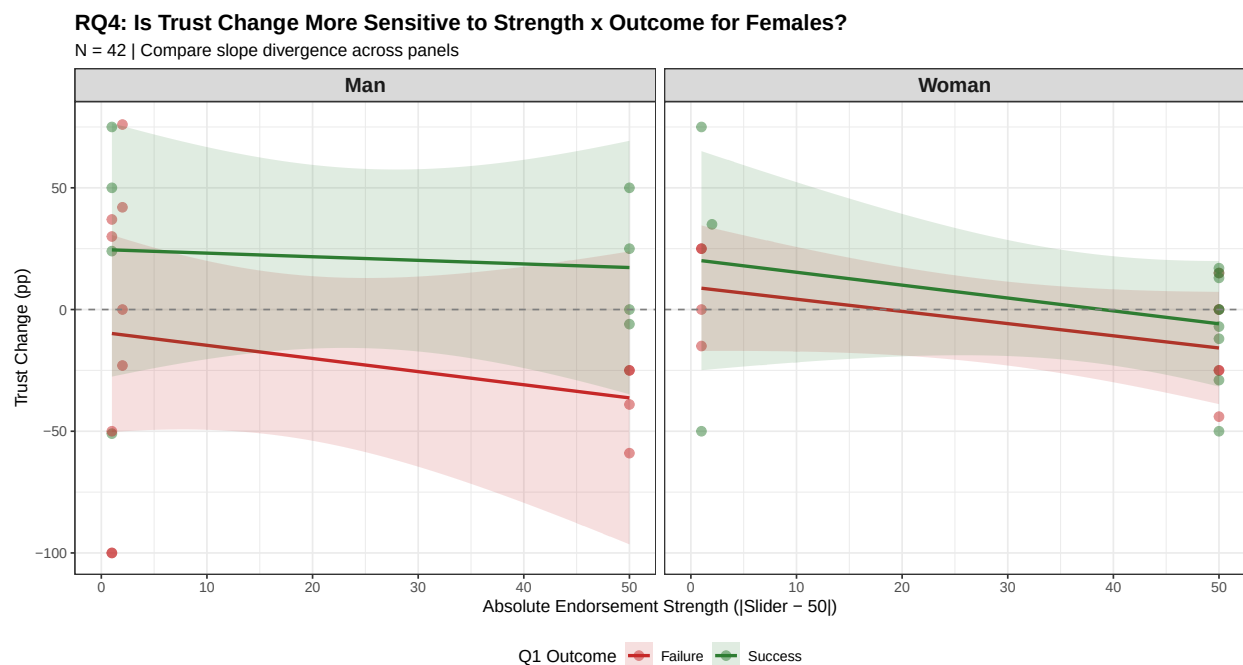
  model_rq4 <- lm(trust_change ~ endorser_gender_f * abs_strength_q1_c * outcome_f,
                  data = df_tc)
  summary(model_rq4)
} else {
  cat("Insufficient data for full 3-way interaction (need >= 15).\n")
  cat("Current N with trust change:", nrow(df_tc), "\n")
}
```

```
##
## Call:
## lm(formula = trust_change ~ endorser_gender_f * abs_strength_q1_c *
##     outcome_f, data = df_tc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -90.153 -21.335   5.854  20.354  86.387
##
## Coefficients:
##                                     Estimate Std. Error
## (Intercept)                       -23.7696    12.7559
## endorser_gender_fWoman              19.6004    18.8884
## abs_strength_q1_c                  -0.5399     0.5162
## outcome_fSuccess                   44.4543    19.5159
## endorser_gender_fWoman:abs_strength_q1_c    0.0389     0.7699
## endorser_gender_fWoman:outcome_fSuccess   -33.8567    27.5649
## abs_strength_q1_c:outcome_fSuccess        0.3920     0.7931
## endorser_gender_fWoman:abs_strength_q1_c:outcome_fSuccess -0.4200     1.1321
##                                     t value Pr(>|t|)
## (Intercept)                       -1.863   0.0711 .
## endorser_gender_fWoman              1.038   0.3067
## abs_strength_q1_c                  -1.046   0.3030
## outcome_fSuccess                   2.278   0.0291 *
## endorser_gender_fWoman:abs_strength_q1_c    0.051   0.9600
## endorser_gender_fWoman:outcome_fSuccess   -1.228   0.2278
## abs_strength_q1_c:outcome_fSuccess        0.494   0.6243
## endorser_gender_fWoman:abs_strength_q1_c:outcome_fSuccess -0.371   0.7129
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 41.72 on 34 degrees of freedom
## Multiple R-squared:  0.1751, Adjusted R-squared:  0.005284
## F-statistic: 1.031 on 7 and 34 DF,  p-value: 0.4279
```

7.1 Visualization

```
if (nrow(df_tc) >= 8) {
  ggplot(df_tc, aes(x = abs_strength_q1, y = trust_change,
                    color = outcome_f, fill = outcome_f)) +
    geom_point(alpha = 0.5, size = 2.5) +
    geom_smooth(method = "lm", se = TRUE, alpha = 0.15) +
    geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
    facet_wrap(~ endorser_gender_f, nrow = 1) +
    scale_color_manual(values = c("Failure" = FAILURE_RED, "Success" = SUCCESS_GRN)) +
    scale_fill_manual(values = c("Failure" = FAILURE_RED, "Success" = SUCCESS_GRN)) +
    scale_x_continuous(breaks = seq(0, 50, 10), limits = c(0, 50)) +
    labs(
      title = "RQ4: Is Trust Change More Sensitive to Strength x Outcome for Females?",
      subtitle = paste0("N = ", nrow(df_tc), " | Compare slope divergence across panels"),
      x = "Absolute Endorsement Strength (|Slider - 50|)",
      y = "Trust Change (pp)", color = "Q1 Outcome", fill = "Q1 Outcome"
    ) +
    theme_bw() +
    theme(
      legend.position = "bottom",
      plot.title = element_text(size = 14, face = "bold"),
      strip.text = element_text(size = 13, face = "bold")
    )
}
```



8. RQ2: Endorsement Strength by Sponsor Gender (Stage 2 Data)

Hypothesis (HQ2): Female sponsors provide **weaker** endorsements (~25% less absolute strength) than male sponsors.

$$\text{AbsStrength}_{ij} = \beta_0 + \beta_1 \text{FemaleSponsor}_i + \beta_2 \text{FemaleProtege}_j + \beta_3 (\text{FemaleSponsor}_i \times \text{FemaleProtege}_j) + \epsilon_{ij}$$

where $\text{AbsStrength} = |\text{Slider} - 50|$ (0 = neutral, 50 = maximally confident).

This analysis uses Stage 2 main study data (N = 200 endorsers, 2,000 trial observations).

8.1 Load Stage 2 Data

```
s2_raw <- read.csv("../output/stage2_main_data.csv", stringsAsFactors = FALSE)
s2_raw <- s2_raw[-c(1, 2), ] # Drop Qualtrics label rows

s2_raw$Finished <- as.numeric(s2_raw$Finished)
s2_raw$StartDate <- as.POSIXct(s2_raw$StartDate, format = "%Y-%m-%d %H:%M:%S")
s2_raw$date <- as.Date(s2_raw$StartDate)

# Filter: not preview, completed, valid PID, main study date
s2 <- s2_raw %>%
  filter(Status != "1", Finished == 1,
         !is.na(PROLIFIC_PID) & trimws(PROLIFIC_PID) != "",
         date >= as.Date("2026-02-12")) %>%
  filter(!is.na(t1_slider_value) & trimws(t1_slider_value) != "")

s2$endorser_gender <- case_when(
  s2$gender == "1" ~ "Man",
  s2$gender == "2" ~ "Woman",
  TRUE ~ "Other"
)

cat("Stage 2 main study endorsers:", nrow(s2), "\n")
```

Stage 2 main study endorsers: 200

```
cat("Gender: Man =", sum(s2$endorser_gender == "Man"),
    "| Woman =", sum(s2$endorser_gender == "Woman"),
    "| Other =", sum(s2$endorser_gender == "Other"), "\n")
```

Gender: Man = 101 | Woman = 95 | Other = 4

```
# Reshape: 10 trials per endorser → long format
s2_long <- data.frame()
for (i in 1:10) {
  p <- paste0("t", i, "_")
  trial_df <- s2 %>%
    select(PROLIFIC_PID, endorser_gender,
```

```

    pair_id = !!paste0(p, "pair_id"),
    category = !!paste0(p, "category"),
    truth = !!paste0(p, "truth"),
    truth_gender = !!paste0(p, "truth_gender"),
    A_gender = !!paste0(p, "A_gender"),
    B_gender = !!paste0(p, "B_gender"),
    slider_value = !!paste0(p, "slider_value"),
    slider_side = !!paste0(p, "slider_side"),
    rt_ms = !!paste0(p, "rt_ms")) %>%
  mutate(trial = i)
s2_long <- rbind(s2_long, trial_df)
}

s2_long$slider_value <- as.numeric(s2_long$slider_value)
s2_long$rt_ms <- as.numeric(s2_long$rt_ms)

# Keep valid trials from Man/Woman endorsers
s2_trials <- s2_long %>%
  filter(!is.na(slider_value), endorser_gender %in% c("Man", "Woman")) %>%
  mutate(
    abs_strength = abs(slider_value - 50),
    endorsed_side = ifelse(slider_value < 50, "A", ifelse(slider_value > 50, "B", "neutral")),
    endorsed_gender = case_when(
      endorsed_side == "A" ~ A_gender,
      endorsed_side == "B" ~ B_gender,
      TRUE ~ NA_character_
    ),
    is_correct = (endorsed_side == truth),
    endorser_gender_f = factor(endorser_gender, levels = c("Man", "Woman")),
    endorsed_gender_f = factor(endorsed_gender, levels = c("Man", "Woman"))
  )

cat("Total trial observations:", nrow(s2_trials), "\n")

## Total trial observations: 1960

cat("Unique endorsers:", n_distinct(s2_trials$PROLIFIC_PID), "\n")

## Unique endorsers: 196

```

8.2 Endorsement Strength by Gender

```

rq2_summary <- s2_trials %>%
  group_by(endorser_gender_f) %>%
  summarise(
    N_endorsers = n_distinct(PROLIFIC_PID),
    N_trials = n(),
    Mean = round(mean(abs_strength), 1),
    SD = round(sd(abs_strength), 1),
    Median = median(abs_strength),
  )

```

```

    .groups = "drop"
  )

  styled_kable(rq2_summary,
    col.names = c("Endorser Gender", "N Endorsers", "N Trials",
                  "Mean |Strength|", "SD", "Median"),
    caption = "Absolute endorsement strength by sponsor gender")

```

Table 13: Absolute endorsement strength by sponsor gender

Endorser Gender	N Endorsers	N Trials	Mean Strength	SD	Median
Man	101	1010	29.1	17.1	30
Woman	95	950	30.8	17.6	35

```

# Also show by endorser x protege gender
rq2_cross <- s2_trials %>%
  filter(!is.na(endorsed_gender_f)) %>%
  group_by(endorser_gender_f, endorsed_gender_f) %>%
  summarise(N = n(), Mean = round(mean(abs_strength), 1),
            SD = round(sd(abs_strength), 1), .groups = "drop")

  styled_kable(rq2_cross,
    col.names = c("Endorser Gender", "Endorsed Candidate Gender",
                  "N", "Mean |Strength|", "SD"),
    caption = "Strength by endorser and endorsed candidate gender")

```

Table 14: Strength by endorser and endorsed candidate gender

Endorser Gender	Endorsed Candidate Gender	N	Mean Strength	SD
Man	Man	458	30.9	14.8
Man	Woman	494	30.9	16.9
Woman	Man	436	32.5	15.6
Woman	Woman	468	32.3	17.5

8.3 RQ2 Model

```

s2_model_data <- s2_trials %>% filter(!is.na(endorsed_gender_f))
model_rq2 <- lm(abs_strength ~ endorser_gender_f * endorsed_gender_f, data = s2_model_data)
summary(model_rq2)

```

```

##
## Call:
## lm(formula = abs_strength ~ endorser_gender_f * endorsed_gender_f,
##     data = s2_model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.521 -14.865   2.733  17.479  19.135

```

```
##
## Coefficients:
##
##              Estimate Std. Error t value
## (Intercept)      30.8646    0.7604  40.591
## endorser_gender_fWoman      1.6560    1.0888   1.521
## endorsed_gender_fWoman      0.0544    1.0556   0.052
## endorser_gender_fWoman:endorsed_gender_fWoman  -0.3080    1.5124  -0.204
##
##              Pr(>|t|)
## (Intercept)      <2e-16 ***
## endorser_gender_fWoman      0.128
## endorsed_gender_fWoman      0.959
## endorser_gender_fWoman:endorsed_gender_fWoman  0.839
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.27 on 1852 degrees of freedom
## Multiple R-squared:  0.002144, Adjusted R-squared:  0.0005275
## F-statistic: 1.326 on 3 and 1852 DF, p-value: 0.2641
```

```
coefs_rq2 <- summary(model_rq2)$coefficients
coef_tbl_rq2 <- data.frame(
  Term = rownames(coefs_rq2),
  Estimate = round(coefs_rq2[, 1], 3),
  SE = round(coefs_rq2[, 2], 3),
  t = round(coefs_rq2[, 3], 3),
  p = round(coefs_rq2[, 4], 4)
)
rownames(coef_tbl_rq2) <- NULL
coef_tbl_rq2$Sig <- ifelse(coef_tbl_rq2$p < 0.05, "*", "")

styled_kable(coef_tbl_rq2, col.names = c("Term", "Estimate", "SE", "t", "p", ""),
  caption = "RQ2 regression coefficients")
```

Table 15: RQ2 regression coefficients

Term	Estimate	SE	t	p	
(Intercept)	30.865	0.760	40.591	0.0000	*
endorser_gender_fWoman	1.656	1.089	1.521	0.1284	
endorsed_gender_fWoman	0.054	1.056	0.052	0.9589	
endorser_gender_fWoman:endorsed_gender_fWoman	-0.308	1.512	-0.204	0.8387	

Interpretation:

- `endorser_gender_fWoman`: Difference in absolute endorsement strength for female vs male sponsors
- `endorsed_gender_fWoman`: Whether endorsing a female candidate changes strength
- Interaction: Whether female sponsors differ specifically when endorsing female candidates

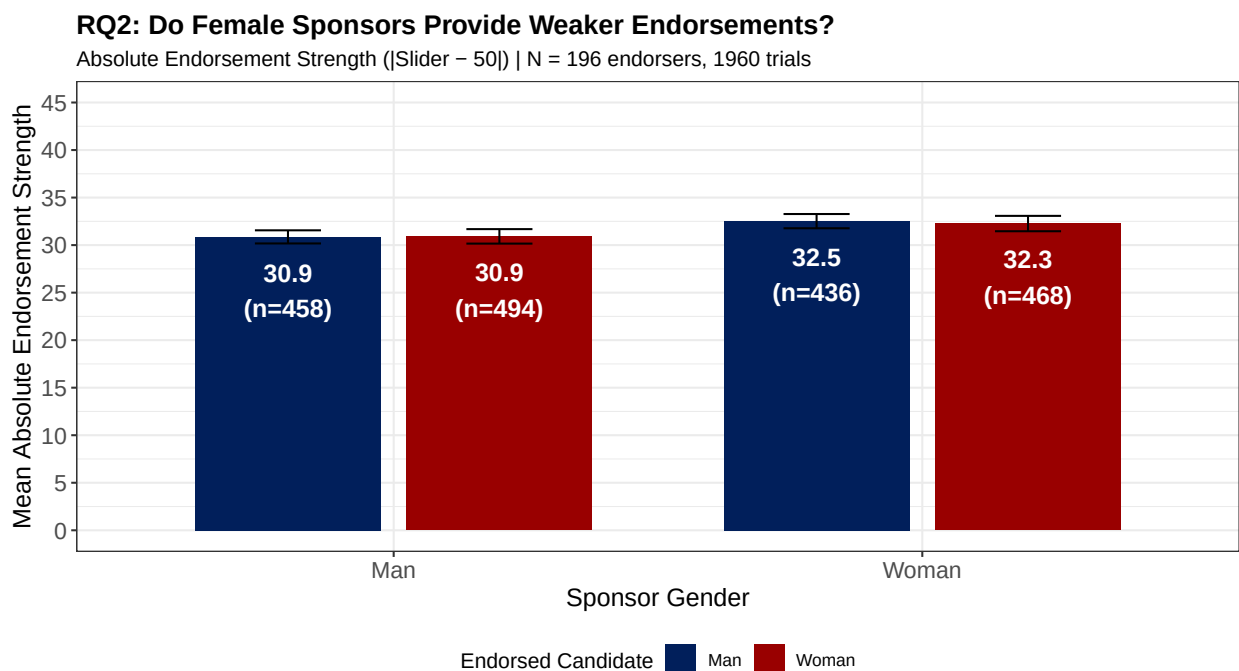
8.4 Visualization


```

rq2_plot <- s2_trials %>%
  filter(!is.na(endorsed_gender_f)) %>%
  group_by(endorser_gender_f, endorsed_gender_f) %>%
  summarise(n = n(), mean_str = mean(abs_strength),
            se = sd(abs_strength) / sqrt(n()), .groups = "drop")

ggplot(rq2_plot,
  aes(x = endorser_gender_f, y = mean_str, fill = endorsed_gender_f)) +
  geom_bar(stat = "identity", position = position_dodge(0.8), width = 0.7) +
  geom_errorbar(aes(ymin = mean_str - se, ymax = mean_str + se),
    position = position_dodge(0.8), width = 0.25) +
  geom_text(aes(label = sprintf("%.1f\n(n=%d)", mean_str, n)),
    position = position_dodge(0.8),
    vjust = 1.5, size = 4.5, color = "white", fontface = "bold") +
  scale_fill_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED),
    name = "Endorsed Candidate") +
  scale_y_continuous(limits = c(0, 45), breaks = seq(0, 45, 5)) +
  labs(
    title = "RQ2: Do Female Sponsors Provide Weaker Endorsements?",
    subtitle = paste0("Absolute Endorsement Strength (|Slider - 50|) | N = ",
      n_distinct(s2_trials$PROLIFIC_PID), " endorsers, ",
      nrow(s2_trials), " trials"),
    x = "Sponsor Gender",
    y = "Mean Absolute Endorsement Strength"
  ) +
  theme_bw() +
  theme(legend.position = "bottom",
    plot.title = element_text(size = 14, face = "bold"),
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 13))

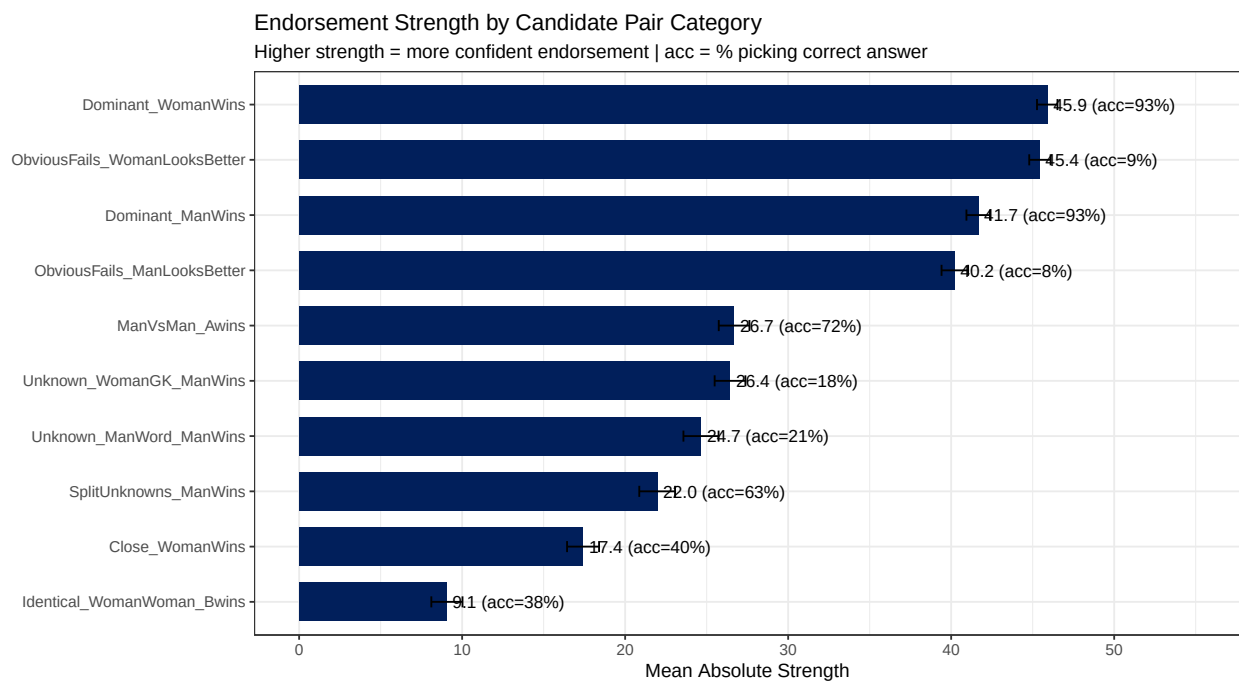
```



8.5 Strength by Candidate Pair Category

```
cat_stats <- s2_trials %>%
  group_by(category) %>%
  summarise(n = n(), mean_str = mean(abs_strength),
            se = sd(abs_strength) / sqrt(n()),
            accuracy = mean(is_correct, na.rm = TRUE), .groups = "drop") %>%
  arrange(desc(mean_str))

ggplot(cat_stats, aes(x = reorder(category, mean_str), y = mean_str)) +
  geom_bar(stat = "identity", fill = PENN_BLUE, width = 0.7) +
  geom_errorbar(aes(ymin = pmax(0, mean_str - se),
                    ymax = mean_str + se), width = 0.2) +
  geom_text(aes(label = sprintf("%.1f (acc=%.0f%%)", mean_str, accuracy * 100)),
            hjust = -0.05, size = 3.5) +
  scale_y_continuous(limits = c(0, 55), breaks = seq(0, 50, 10)) +
  coord_flip() +
  labs(title = "Endorsement Strength by Candidate Pair Category",
       subtitle = "Higher strength = more confident endorsement | acc = % picking correct answer",
       x = NULL, y = "Mean Absolute Strength") +
  theme_bw()
```



8.6 Endorser Accuracy

```
acc_tbl <- s2_trials %>%
  filter(endorsed_side != "neutral") %>%
  group_by(endorser_gender_f) %>%
  summarise(
```

```

N = n(),
Correct = sum(is_correct, na.rm = TRUE),
Accuracy = paste0(round(mean(is_correct, na.rm = TRUE) * 100, 1), "%"),
.groups = "drop"
)

styled_kable(acc_tbl, col.names = c("Endorser Gender", "N Decisions", "N Correct", "Accuracy"),
  caption = "Endorser accuracy (picking the objectively better candidate)")

```

Table 16: Endorser accuracy (picking the objectively better candidate)

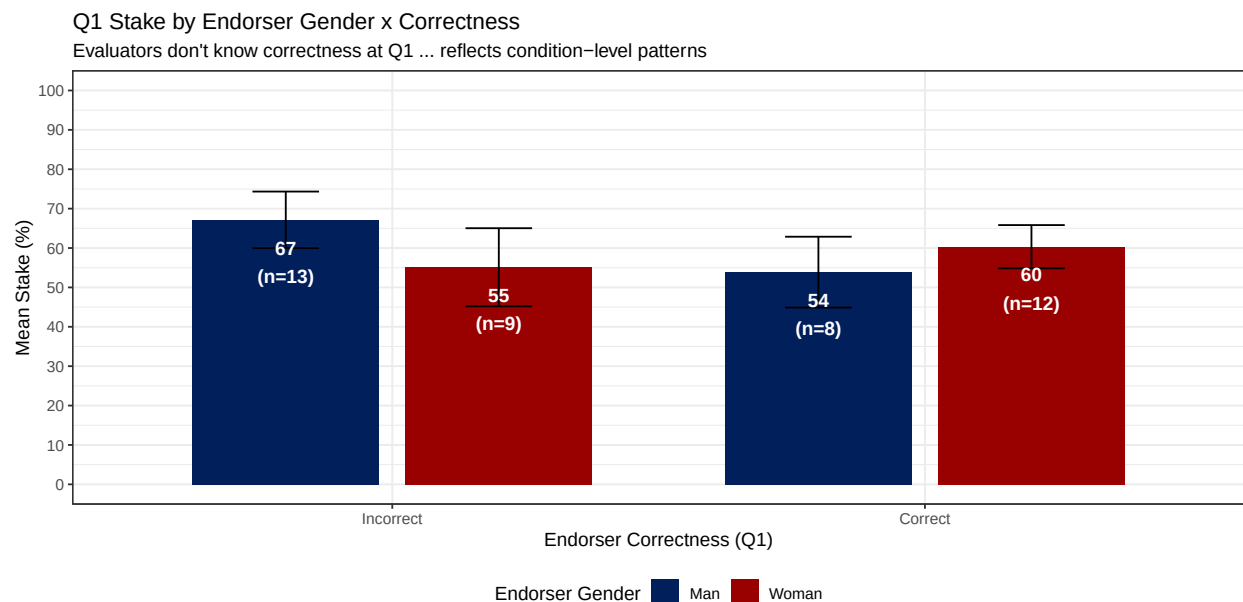
Endorser Gender	N Decisions	N Correct	Accuracy
Man	952	458	48.1%
Woman	904	435	48.1%

9. Descriptive Patterns (Full Factorial, Stage 3)

9.1 Gender x Correctness on Q1 Stake

```
gc_stats <- df %>%
  group_by(endorser_gender_f, correctness_f) %>%
  summarise(n = n(), mean_stake = mean(stake_percent_q1),
            se = sd(stake_percent_q1) / sqrt(n()), .groups = "drop")

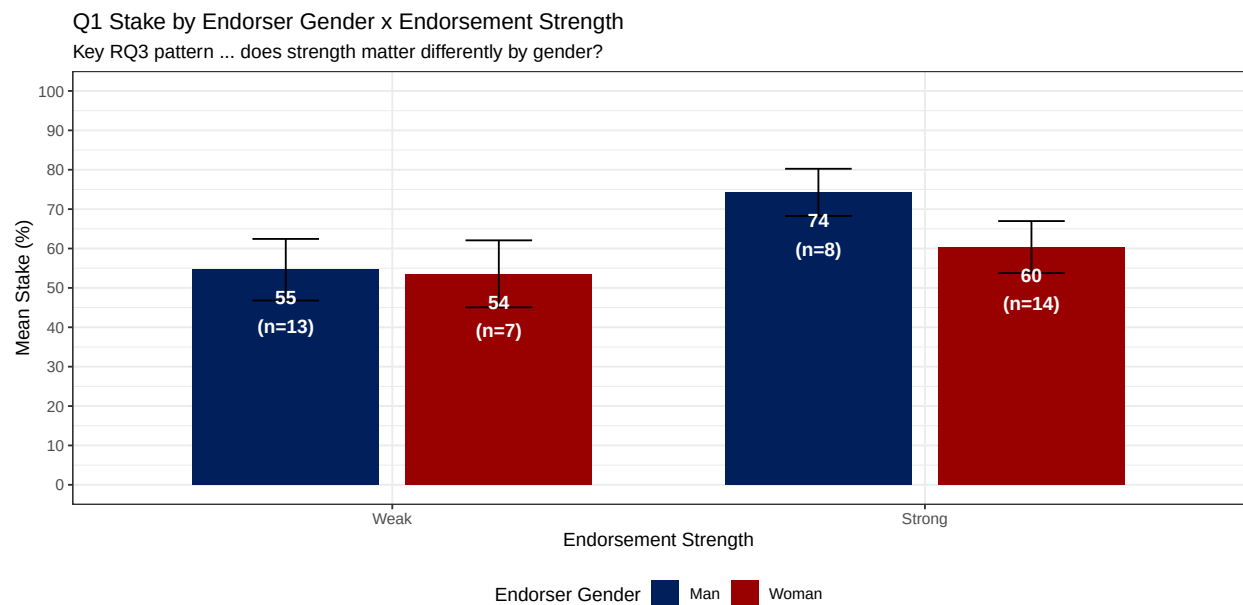
ggplot(gc_stats, aes(x = correctness_f, y = mean_stake, fill = endorser_gender_f)) +
  geom_bar(stat = "identity", position = position_dodge(0.8), width = 0.7) +
  geom_errorbar(aes(ymin = mean_stake - se, ymax = mean_stake + se),
               position = position_dodge(0.8), width = 0.25) +
  geom_text(aes(label = sprintf("%.0f\n(n=%d)", mean_stake, n)),
            position = position_dodge(0.8), vjust = 1.5,
            color = "white", fontface = "bold", size = 4) +
  scale_fill_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED)) +
  scale_y_continuous(limits = c(0, 100), breaks = seq(0, 100, 10)) +
  labs(title = "Q1 Stake by Endorser Gender x Correctness",
       subtitle = "Evaluators don't know correctness at Q1 - reflects condition-level patterns",
       x = "Endorser Correctness (Q1)", y = "Mean Stake (%)", fill = "Endorser Gender") +
  theme_bw() + theme(legend.position = "bottom")
```



9.2 Gender x Strength on Q1 Stake

```
gs_stats <- df %>%
  group_by(endorser_gender_f, strength_f) %>%
  summarise(n = n(), mean_stake = mean(stake_percent_q1),
            se = sd(stake_percent_q1) / sqrt(n()), .groups = "drop")
```

```
ggplot(gs_stats, aes(x = strength_f, y = mean_stake, fill = endorser_gender_f)) +
  geom_bar(stat = "identity", position = position_dodge(0.8), width = 0.7) +
  geom_errorbar(aes(ymin = mean_stake - se, ymax = mean_stake + se),
    position = position_dodge(0.8), width = 0.25) +
  geom_text(aes(label = sprintf("%.0f\n(n=%d)", mean_stake, n)),
    position = position_dodge(0.8), vjust = 1.5,
    color = "white", fontface = "bold", size = 4) +
  scale_fill_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED)) +
  scale_y_continuous(limits = c(0, 100), breaks = seq(0, 100, 10)) +
  labs(title = "Q1 Stake by Endorser Gender x Endorsement Strength",
    subtitle = "Key RQ3 pattern - does strength matter differently by gender?",
    x = "Endorsement Strength", y = "Mean Stake (%)", fill = "Endorser Gender") +
  theme_bw() + theme(legend.position = "bottom")
```



9.3 Full 2x2x2 Heatmap

```
full_stats <- df %>%
  group_by(endorser_gender_f, correctness_f, strength_f) %>%
  summarise(n = n(), mean_stake = mean(stake_percent_q1), .groups = "drop")

all_combos <- expand_grid(
  endorser_gender_f = factor(c("Man", "Woman"), levels = c("Man", "Woman")),
  correctness_f = factor(c("Incorrect", "Correct"), levels = c("Incorrect", "Correct")),
  strength_f = factor(c("Weak", "Strong"), levels = c("Weak", "Strong"))
)

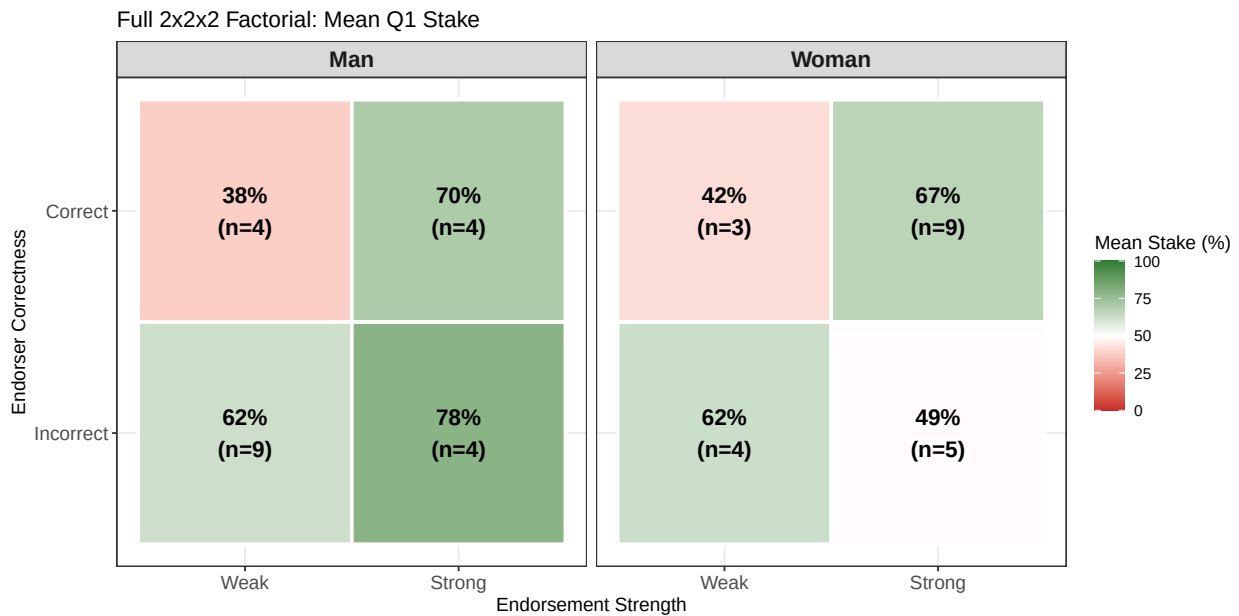
full_stats <- all_combos %>%
  left_join(full_stats, by = c("endorser_gender_f", "correctness_f", "strength_f"))

ggplot(full_stats, aes(x = strength_f, y = correctness_f, fill = mean_stake)) +
  geom_tile(color = "white", size = 1) +
```

```

geom_text(aes(label = ifelse(is.na(n), "---",
                             sprintf("%.0f%%\n(n=%d)", mean_stake, n))),
          size = 4.5, fontface = "bold") +
facet_wrap(~ endorser_gender_f) +
scale_fill_gradient2(low = FAILURE_RED, mid = "white", high = SUCCESS_GRN,
                     midpoint = 50, na.value = "gray90",
                     limits = c(0, 100), name = "Mean Stake (%)") +
labs(title = "Full 2x2x2 Factorial: Mean Q1 Stake",
     x = "Endorsement Strength", y = "Endorser Correctness") +
theme_bw() +
theme(strip.text = element_text(size = 13, face = "bold"),
      axis.text = element_text(size = 11))

```



9.4 Trust Change Heatmap (for participants with Q2)

```

if (nrow(df_tc) >= 4) {
  tc_stats <- df_tc %>%
    group_by(endorser_gender_f, outcome_f, strength_f) %>%
    summarise(n = n(), mean_tc = mean(trust_change), .groups = "drop")

  tc_combos <- expand_grid(
    endorser_gender_f = factor(c("Man", "Woman"), levels = c("Man", "Woman")),
    outcome_f = factor(c("Failure", "Success"), levels = c("Failure", "Success")),
    strength_f = factor(c("Weak", "Strong"), levels = c("Weak", "Strong"))
  )
  tc_stats <- tc_combos %>%
    left_join(tc_stats, by = c("endorser_gender_f", "outcome_f", "strength_f"))

  ggplot(tc_stats, aes(x = strength_f, y = outcome_f, fill = mean_tc)) +
    geom_tile(color = "white", size = 1) +
    geom_text(aes(label = ifelse(is.na(n), "---",

```

```

        sprintf("%+.0f\n(n=%d)", mean_tc, n))),
        size = 4.5, fontface = "bold") +
facet_wrap(~ endorser_gender_f) +
scale_fill_gradient2(low = FAILURE_RED, mid = "white", high = SUCCESS_GRN,
                     midpoint = 0, na.value = "gray90",
                     name = "Mean Trust\nChange (pp)") +
labs(title = "Trust Change by Gender x Outcome x Strength",
     subtitle = paste0("N with Q2 data = ", nrow(df_tc)),
     x = "Endorsement Strength", y = "Q1 Outcome") +
theme_bw() +
theme(strip.text = element_text(size = 13, face = "bold"))
}

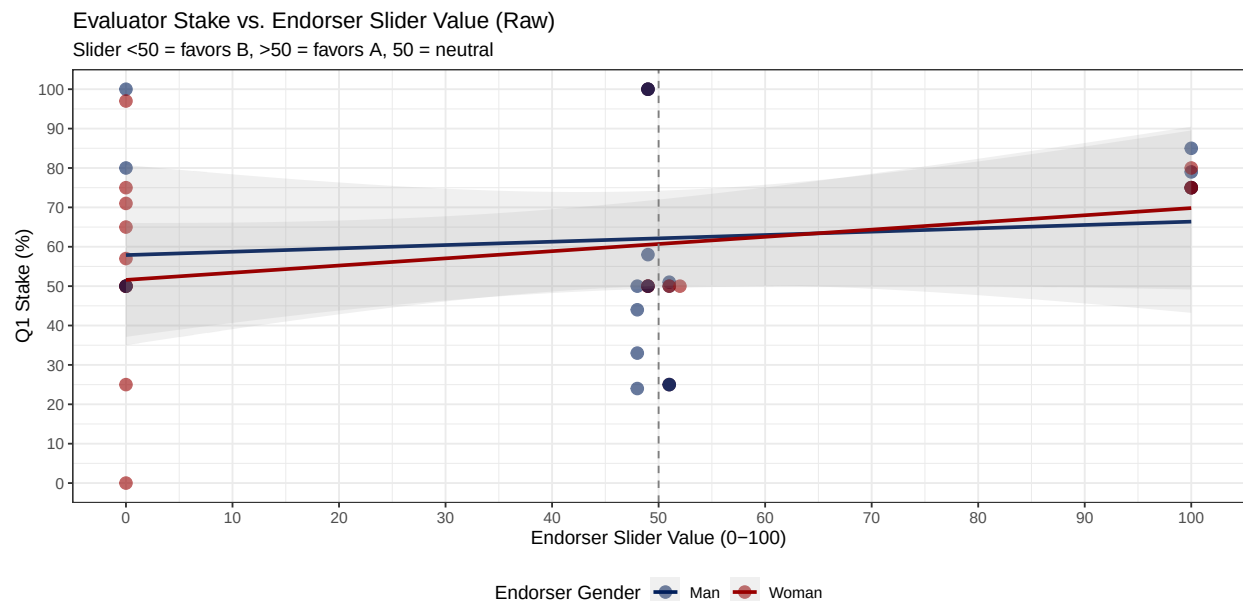
```



10. Exploratory Analyses

10.1 Endorser Confidence and Evaluator Stake

```
ggplot(df, aes(x = endorser_slider_value_q1, y = stake_percent_q1,
               color = endorser_gender_f)) +
  geom_point(alpha = 0.6, size = 3) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15) +
  geom_vline(xintercept = 50, linetype = "dashed", color = "gray50") +
  scale_color_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED)) +
  scale_x_continuous(breaks = seq(0, 100, 10)) +
  scale_y_continuous(limits = c(0, 100), breaks = seq(0, 100, 10)) +
  labs(title = "Evaluator Stake vs. Endorser Slider Value (Raw)",
       subtitle = "Slider <50 = favors B, >50 = favors A, 50 = neutral",
       x = "Endorser Slider Value (0-100)", y = "Q1 Stake (%)",
       color = "Endorser Gender") +
  theme_bw() + theme(legend.position = "bottom")
```



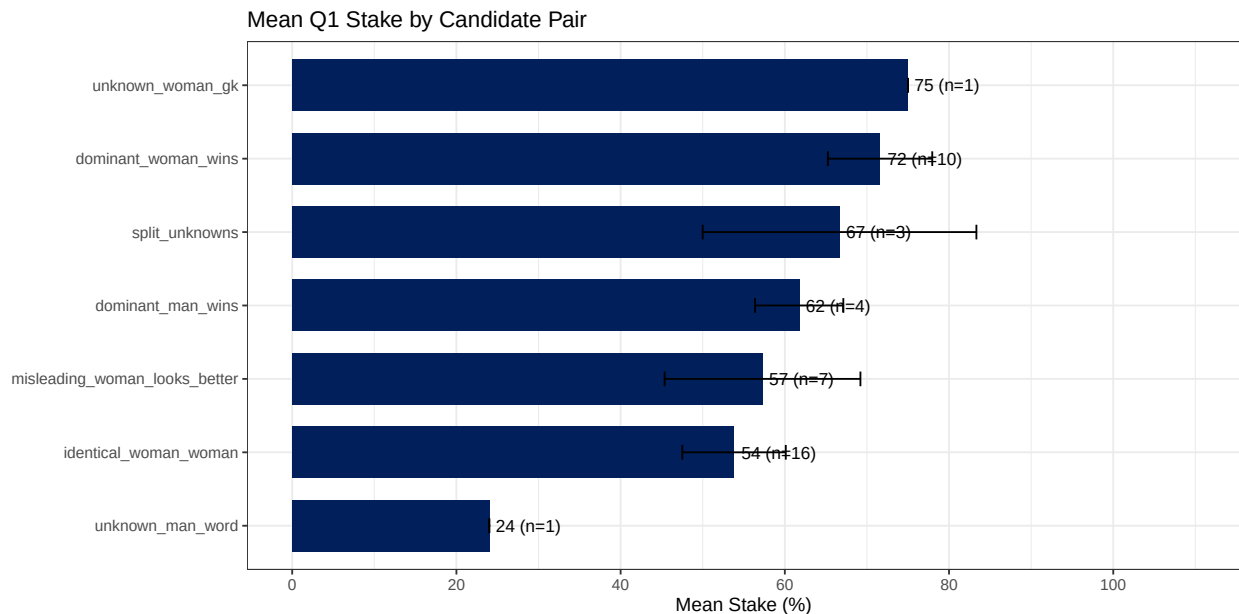
10.2 Q1 Stake by Candidate Pair

```
pair_stats <- df %>%
  group_by(stage3_q1_pair_id) %>%
  summarise(n = n(), mean_stake = mean(stake_percent_q1),
            se = ifelse(n() > 1, sd(stake_percent_q1) / sqrt(n()), 0),
            .groups = "drop") %>%
  filter(n >= 1)

ggplot(pair_stats, aes(x = reorder(stage3_q1_pair_id, mean_stake), y = mean_stake)) +
  geom_bar(stat = "identity", fill = PENN_BLUE, width = 0.7) +
```



```
geom_errorbar(aes(ymin = pmax(0, mean_stake - se),
                  ymax = pmin(100, mean_stake + se)), width = 0.2) +
geom_text(aes(label = sprintf("%.0f (n=%d)", mean_stake, n)),
          hjust = -0.1, size = 3.5) +
scale_y_continuous(limits = c(0, 110), breaks = seq(0, 100, 20)) +
coord_flip() +
labs(title = "Mean Q1 Stake by Candidate Pair", x = NULL, y = "Mean Stake (%)") +
theme_bw()
```



10.3 Response Time Analysis

```
df_rt <- df %>% filter(rt_ms_stage3_q1 > 0)

if (nrow(df_rt) > 0) {
  df_rt <- df_rt %>%
    mutate(rt_sec_q1 = rt_ms_stage3_q1 / 1000,
           rt_sec_q2 = rt_ms_stage3_q2 / 1000)

  cat("Response time stats (seconds):\n")
  cat("  Q1: M =", round(mean(df_rt$rt_sec_q1, na.rm = TRUE), 1),
      "| Median =", round(median(df_rt$rt_sec_q1, na.rm = TRUE), 1),
      "| SD =", round(sd(df_rt$rt_sec_q1, na.rm = TRUE), 1), "\n")
  if (sum(df_rt$rt_sec_q2 > 0, na.rm = TRUE) > 0) {
    cat("  Q2: M =", round(mean(df_rt$rt_sec_q2[df_rt$rt_sec_q2 > 0], na.rm = TRUE), 1),
        "| Median =", round(median(df_rt$rt_sec_q2[df_rt$rt_sec_q2 > 0], na.rm = TRUE), 1), "\n")
  }

  ggplot(df_rt, aes(x = rt_sec_q1, y = stake_percent_q1, color = endorser_gender_f)) +
    geom_point(alpha = 0.6, size = 3) +
    geom_smooth(method = "lm", se = TRUE, alpha = 0.15) +
    scale_color_manual(values = c("Man" = PENN_BLUE, "Woman" = PENN_RED)) +
```

```

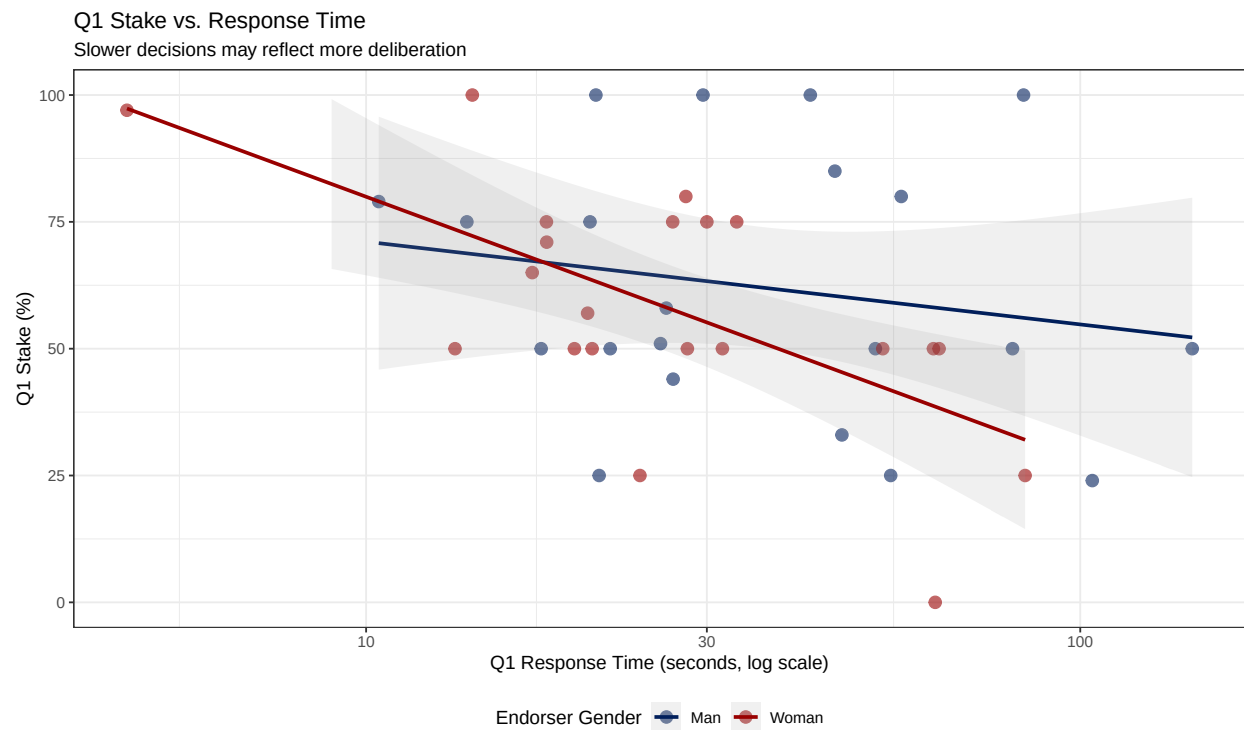
scale_x_continuous(trans = "log10") +
scale_y_continuous(limits = c(0, 100)) +
labs(title = "Q1 Stake vs. Response Time",
      subtitle = "Slower decisions may reflect more deliberation",
      x = "Q1 Response Time (seconds, log scale)", y = "Q1 Stake (%)",
      color = "Endorser Gender") +
theme_bw() + theme(legend.position = "bottom")
} else {
  cat("No response time data available.\n")
}

```

```

## Response time stats (seconds):
##   Q1: M = 38.4 | Median = 27.5 | SD = 28.3
##   Q2: M = 16.5 | Median = 12.1

```



11. Preliminary Effect Sizes & Power

```
man_stakes <- df$stake_percent_q1[df$endorser_gender_code == "Man"]
woman_stakes <- df$stake_percent_q1[df$endorser_gender_code == "Woman"]

pooled_sd <- sqrt(((length(man_stakes) - 1) * var(man_stakes) +
                    (length(woman_stakes) - 1) * var(woman_stakes)) /
                  (length(man_stakes) + length(woman_stakes) - 2))
cohens_d <- (mean(man_stakes) - mean(woman_stakes)) / pooled_sd

cat("=== Preliminary Effect Sizes (Q1 Stake) ===\n\n")

## === Preliminary Effect Sizes (Q1 Stake) ===

cat("Gender effect:\n")

## Gender effect:

cat("  Man mean:", round(mean(man_stakes), 1), "(n=", length(man_stakes), ")\n")

##   Man mean: 62.1 (n= 21 )

cat("  Woman mean:", round(mean(woman_stakes), 1), "(n=", length(woman_stakes), ")\n")

##   Woman mean: 58.1 (n= 21 )

cat("  Cohen's d:", round(cohens_d, 3), "\n\n")

##   Cohen's d: 0.161

r2 <- summary(model_rq3)$r.squared
cat("RQ3 model R-squared:", round(r2, 3), "\n")

## RQ3 model R-squared: 0.09

cat("RQ3 model Adj. R-squared:", round(summary(model_rq3)$adj.r.squared, 3), "\n\n")

## RQ3 model Adj. R-squared: 0.018

if (nrow(df_tc) >= 10 && exists("model_rq1")) {
  cat("=== Preliminary Effect Sizes (Trust Change) ===\n\n")
  cat("RQ1 model R-squared:", round(summary(model_rq1)$r.squared, 3), "\n")
  cat("RQ1 model Adj. R-squared:", round(summary(model_rq1)$adj.r.squared, 3), "\n")
}

## === Preliminary Effect Sizes (Trust Change) ===
##
## RQ1 model R-squared: 0.108
## RQ1 model Adj. R-squared: 0.037
```

```
cat("\n=== Power Note ===\n")
```

```
##
```

```
## === Power Note ===
```

```
cat("Current N =", nrow(df), "(Q1) /", nrow(df_tc), "(trust change)\n")
```

```
## Current N = 42 (Q1) / 42 (trust change)
```

```
cat("Target main study N = 600 evaluators\n")
```

```
## Target main study N = 600 evaluators
```

```
cat("At current N, detectable effects are very large (d > ~0.8).\n")
```

```
## At current N, detectable effects are very large (d > ~0.8).
```

```
cat("Primary value: verify mechanics, check data quality, observe directional patterns.\n")
```

```
## Primary value: verify mechanics, check data quality, observe directional patterns.
```

12. Summary

Sample & Data Quality

```
summary_tbl <- data.frame(  
  Metric = c(  
    "Total Prolific completions",  
    "Passed comprehension check (Q42)",  
    "Final analytic sample (all filters)",  
    "With Q2 stake (trust change available)",  
    "With response times",  
    "Conditions covered (of 8)",  
    "Median completion time"  
  ),  
  Value = c(  
    nrow(d3),  
    nrow(d4),  
    nrow(df),  
    nrow(df_tc),  
    n_rt,  
    n_conds,  
    paste0(round(median(df$duration_min, na.rm = TRUE), 1), " min")  
  )  
)  
styled_kable(summary_tbl)
```

Metric	Value
Total Prolific completions	130
Passed comprehension check (Q42)	42
Final analytic sample (all filters)	42
With Q2 stake (trust change available)	42
With response times	42
Conditions covered (of 8)	8
Median completion time	5.9 min

Early Directional Patterns

```
cat("These are DESCRIPTIVE pilot patterns - not confirmatory tests.\n\n")
```

```
## These are DESCRIPTIVE pilot patterns - not confirmatory tests.
```

```
cat("1. GENDER on Q1 stake: Man M =", round(mean(man_stakes), 1),  
    "vs. Woman M =", round(mean(woman_stakes), 1),  
    "(d =", round(cohens_d, 2), ")\n")
```

```
## 1. GENDER on Q1 stake: Man M = 62.1 vs. Woman M = 58.1 (d = 0.16 )
```

```

strong_stakes <- df$stake_percent_q1[df$endorser_strong == 1]
weak_stakes   <- df$stake_percent_q1[df$endorser_strong == 0]
if (length(weak_stakes) > 1 && length(strong_stakes) > 1) {
  cat("2. STRENGTH on Q1 stake: Strong M =", round(mean(strong_stakes), 1),
      "vs. Weak M =", round(mean(weak_stakes), 1), "\n")
}

```

2. STRENGTH on Q1 stake: Strong M = 65.4 vs. Weak M = 54.2

```

if (nrow(df_tc) > 0) {
  cat("3. TRUST CHANGE overall: M =", round(mean(df_tc$trust_change), 1),
      "(SD =", round(sd(df_tc$trust_change), 1), ") \n")
}

```

3. TRUST CHANGE overall: M = -2.5 (SD = 41.8)

Analysis generated 2026-02-12 16:39:40.058431 / Stage 3 Pilot (Study 698e27db580428e78f8e9ff1) / Sponsorship Network